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CONVERGENCE ANALYSIS FOR A CLASS OF HIGH-ORDER SEMI-LAGRANGIAN ADVECTION SCHEMES*

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Abstract. The convergence properties of a class of high-order semi-Lagrangian schemes for pure advection equations are studied here in the framework of the theory of viscosity solutions. We review the general convergence results for discrete-time approximation schemes belonging to that class and we prove some a priori estimates in L^∞ and L^2 for the rate of convergence of fully discrete schemes. We prove then that a careful coupling of time and space discretizations can allow large time steps in the numerical integration still preserving the accuracy of the solutions. Several examples of schemes and numerical tests are presented.

Key words. hyperbolic equations, method of characteristics, semi-Lagrangian schemes, high-order schemes, convergence, stability.

AMS subject classifications. 65M25, 65M12, 65N06

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Introduction. We will examine in this paper a class of semi-Lagrangian approximation schemes for the advection equation. This class of methods has been introduced at the beginning of the eighties (see Robert [R]) and since then has been extensively used in the numerical simulation of models for weather forecast, oceanography, and for more general problems in fluid dynamics (see Staniforth and Coté [SC] for an up-to-date review, Smolarkiewicz and Pudykiewicz [SP] and Smolarkiewicz and Grell [SG] for more recent developments). The basic idea of the semi-Lagrangian method is to reconstruct the solution over a grid integrating numerically the equation along the characteristics starting from any grid point x_i. This means that the solution in x_i will be computed coupling a numerical method for ODEs (to recover the upwind points with respect to the grid nodes) with an interpolation formula (to recover the value of the solution in those points). From this point of view, this method can be interpreted as a discrete and *local* version of the classical method of characteristics (see, e.g., Forsythe and Wasow [FW] and Richtmeyer and Morton [RM] for the numerical applications of the *global* method of characteristics).

Despite the large number of papers reporting numerical results obtained via semi-Lagrangian methods we are not aware of theoretical a priori estimates which can be applied to such methods in the general situation when the solution is just continuous. In fact, theoretical results about the order of convergence of such methods mainly rely on the truncation error computed by means of Taylor expansion and require the strong regularity of the solution. In this paper we will always consider a solution to be a weak one in the viscosity sense (see Crandall–Ishii–Lions [CIL] for an up-to-date presentation of this theory). Roughly speaking, this means that we deal with particular Lipschitz continuous solutions. Our goal is to establish to which extent the semi-Lagrangian approximation can produce high-order numerical schemes for linear

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first-order (advection) equations of both evolutive and stationary types. We will focus our attention on the pure advection equations since it is well known that their numerical solution is crucial in many applications in fluid dynamics where the first-order term dominates on higher-order terms in a single equation (convection-diffusion problems with low viscosity) and/or as part of a system (Navier–Stokes, Euler). We should note, however, that this approach can also be applied to other problems such as the study of coupled sets of equations [SC], monotone interpolation [SG], numerical schemes in spheric geometries for weather prediction [SC], or for nonlinear Hamilton–Jacobi equations [FF].

Let us consider, as our first model, the evolutive problem,

$$\text{(EP)} \qquad \begin{cases} \dfrac{\partial}{\partial t} v(x,t) = \lambda v(x,t) + f(x) \cdot \nabla v(x,t) + g(x) & \text{in } \Omega \times [0,T], \\ v(x,0) = v_0(x) \end{cases}$$

where $\lambda \in \mathbb{R}$, $t \in [0,T]$. When $\lambda < 0$, the solution $v(x,t)$ of (EP) converges for t going to $+\infty$ to the solution $v(x)$ of the stationary problem

$$\text{(SP)} \qquad \lambda v(x) + f(x) \cdot \nabla v(x) + g(x) = 0 \qquad \text{in } \Omega,$$

which will be our second model problem. We will derive the semi-Lagrangian schemes from the representation formulae for the solutions of (EP), (SP). This point of view is also useful because it can be applied to nonlinear problems. In particular, the technique used here for the time-discretization has been extensively studied in [FF] for a nonlinear stationary Hamilton–Jacobi equation $u + H(x, \nabla u) = 0$ where the Hamiltonian H is convex with respect to ∇u. The general theory developed in this paper can also deal with more complicated situations in which the parameter λ depends on the state variable x and/or the terms f and g depend on t (see Remark 4).

Following this approach, the numerical scheme can be interpreted as a discrete version of the representation formula. This discretization naturally arises coupling a one-step scheme for the ODE describing the characteristics with a quadrature formula. Several numerical methods for ODEs and quadrature formulae can be coupled obtaining a class of high-order schemes for (EP), (SP).

In the space discretization either structured or unstructured grids may be used. In both cases the grid need not be “adapted” to the vector field f. We will first study (EP) and (SP) for $\Omega = \mathbb{R}^N$ to obtain results on the convergence of the algorithm without boundary conditions. We will then set the problem in a bounded domain Ω and show how to treat Dirichlet or periodic boundary conditions numerically.

We should also mention that many other methods have been developed for advection problems: finite differences, stream-line diffusions, finite elements, and spectral techniques. A blend of the method of characteristics with finite elements and finite differences has been studied in recent years by Douglas and Russell [DR], Hasbani, Livne, and Bercovier [HLB], Morton, Priestley, and Süli [MPS], and Pironneau [P] mainly for the numerical solution of advection-diffusion and Navier–Stokes equations. We refer to Süli and Ware [SW] for a coupling between the method of characteristics and spectral methods applied to the numerical solutions of first-order hyperbolic equations. A more direct and effective approach is based on TVD schemes, in particular, ENO schemes (see the papers by Osher and Shu [OS, SO1, SO2], Leveque [L], and references therein).

Our contribution here is to develop a general theory for a class of semi-Lagrangian schemes for linear first-order problems. The main results are a general convergence theorem, some global a priori estimates for a suitable norm of the error, and pointwise estimates away from discontinuities (see Theorem 4.2). Such estimates and several numerical experiments show that an interesting feature of the above schemes is to allow (in certain conditions, see Remark 10) large time steps still having an accurate numerical solution. Moreover, the qualitative behavior of the schemes considered here is similar to the above mentioned ENO schemes. This indication also seems to be confirmed by the comparison of our results for the rotating cone problem in section 7 with the tests reported in Tamamidis and Assanis [TA].

The outline of the paper is as follows. Sections 2 and 3 deal with time discretization for (EP) and (SP), respectively. In sections 4 and 6 we will treat space discretization, proving convergence and a priori estimates for the fully discrete schemes. Stability is analyzed in section 5. Section 7 is devoted to the introduction of periodic and Dirichlet boundary conditions in the scheme. Finally, in section 8 we present the results of some numerical tests in $\mathbb{R}^1$ and $\mathbb{R}^2$.

1. Assumptions and basic results. We assume that the functions $f : \mathbb{R}^N \to \mathbb{R}^N$ and $g : \mathbb{R}^N \to \mathbb{R}$ are bounded and (at least) Lipschitz continuous, that is,

$$\|f\|_\infty \le M_f \ , \quad |f(x_1) - f(x_2)| \le L_f|x_1 - x_2|, \tag{h1a}$$

$$\|g\|_\infty \le M_g \ , \quad |g(x_1) - g(x_2)| \le L_g|x_1 - x_2|, \tag{h1b}$$

for any $x_1, x_2 \in \mathbb{R}^N$. Here and in the sequel, we will denote as usual by $\|\cdot\|_\infty$ the norm in L^∞ (it will be clear from the context if it refers to $L^\infty(\mathbb{R}^N)$ or to $L^\infty(\Omega)$) and by $\|\cdot\|_2$ the norm in $L^2(\Omega)$.

As we said in the introduction, we will work in the framework of viscosity solutions (see [Li] and [CIL]). It is known that, under the assumptions (h1), (EP) has a unique uniformly continuous viscosity solution in $\mathbb{R}^N \times [0, T]$ provided $v_0(x)$ is continuous. When (EP) is set in a domain $\Omega \times [0, T]$ one should add a Dirichlet boundary condition on Γ_{in}, the inflow part of $\partial\Omega$:

$$v(x, t) = b(x), \qquad\qquad \forall x \in \Gamma_{in}, t \in [0, T],$$

and in that situation discontinuities can appear depending on the regularity of b (a similar theory applies to (SP)). The notion of viscosity solution has also been extended to include discontinuous solutions (see the references in [CIL]). We just note that the schemes discussed here are robust enough to give reasonably accurate results also for discontinuous solutions, at least in a local sense (see Theorem 4.2 and section 8).

For completeness, we start our presentation by the representation formula for the solutions of (EP) and (SP). Let $y(\cdot), y^-(\cdot) \in \mathbb{R}^N$ be, respectively, the (unique) solutions of the initial value problems:

$$\begin{cases} \dot{y}(s) = f(y(s)), \\ y(0) = x, \end{cases} \tag{1.1}$$

$$\begin{cases} \dot{y}^-(s) = -f(y^-(s)), \\ y^-(0) = x \end{cases} \tag{1.2}$$

(clearly, $y^-(t) = y(-t)$). If we write (EP) at a point $(y^-(s), s)$ and multiply by $e^{-\lambda s}$, assuming v regular we obtain

$$e^{-\lambda s}\frac{\partial}{\partial s}v(y^-(s), s) - \lambda e^{-\lambda s}v(y^-(s), s) - e^{-\lambda s}f(y^-(s)) \cdot \nabla v(y^-(s), s) = e^{-\lambda s}g(y^-(s)),$$

which may be rewritten as

$$\frac{d}{ds}\left[e^{-\lambda s}v(y^-(s),s)\right] = e^{-\lambda s}g(y^-(s)). \tag{1.3}$$

Integrating (1.3) over $[-t,0]$, since g does not depend on t, we get

$$v(x,t) = \int_{-t}^{0} e^{-\lambda s}g(y^-(s))ds + e^{\lambda t}v_0(y^-(-t))$$

and, hence,

$$v(x,t) = \int_0^t e^{\lambda s}g(y(s))ds + e^{\lambda t}v_0(y(t)). \tag{1.4}$$

In particular, when $\lambda < 0$ and v_0 is bounded, the solution $v(x)$ of (SP) can be written as

$$v(x) = \int_0^\infty e^{\lambda s}g(y(s))ds. \tag{1.5}$$

On the other hand, when $\lambda = 0$ and $g(\cdot) \equiv 0$, we can write the solution $v(x,t)$ of (EP) as

$$v(x,t) = v_0(y(t)), \tag{1.6}$$

which shows, as it is well known, that the trajectories of dynamical system (1.2) are characteristics of (EP) (i.e., $v(x,t)$ is constant along these curves).

The representation formulae (1.4), (1.5) allow us to derive two relationships which might also be seen as particular cases of the dynamic programming principle. Setting a time step $h > 0$, we can obtain from (1.4)

$$v(x,t) = \int_0^h e^{\lambda s}g(y(s))ds + e^{\lambda h}v(y(h),t-h). \tag{1.7}$$

On the other hand, splitting the integral in (1.5) in the sum of the two integrals on $[0,h]$ and $[h,+\infty]$, we get

$$v(x) = \int_0^h e^{\lambda s}g(y(s))ds + e^{\lambda h}v(y(h)), \tag{1.8}$$

which coincides with (1.7) when applied to the limit solution $v(x)$.

This heuristic derivation of the representation formulae (1.4) and (1.5) can be made rigorous in the framework of viscosity solutions (see [Li]).

2. Time discretization for the evolution equation. We will first treat the general case (EP). In order to obtain a discrete-time approximation, we will use an explicit Runge–Kutta discretization of order $p \geq 1$. It will be convenient to consider the dynamical system

$$\begin{pmatrix} \dot{y}(s) \\ \dot{\gamma}(s) \end{pmatrix} = \begin{pmatrix} f(y(s)) \\ e^{\lambda s}g(y(s)) \end{pmatrix} \tag{2.1}$$

with the initial conditions $y(0) = x$, $\gamma(0) = 0$. This form allows one to use the same one-step scheme for the discretization of both the ODE (1.1) and the integral representation (1.4). More precisely, we have

$$\begin{pmatrix} y_1 \\ \gamma_1 \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix} + h\begin{pmatrix} \Phi_f(x) \\ \Phi_g(x) \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix} + h\Phi(x) \tag{2.2}$$

where we have referred to the first time step and splitted the function Φ according to (2.1). Now, it turns out from (2.2) that y_1 is an approximation of $y(h)$, whereas γ_1 approximates the integral in (1.7).

We will assume consistency:

$$\lim_{h\to 0} \Phi_f(x,h) = f(x), \tag{h2a}$$

$$\lim_{h\to 0} \Phi_g(x,h) = g(x), \tag{h2b}$$

and Lipschitz continuity of Φ:

$$|\Phi(x_1,h) - \Phi(x_2,h)| \le L_\Phi |x_1 - x_2| \tag{h2c}$$

for $0 < h < h_0$. These assumptions are always satisfied by Runge–Kutta schemes. Even if the function Φ depends in general on h, we will use the simplified notation $\Phi(x) := \Phi(x,h)$ whenever we do not need to take into account this dependence.

Using the one-step approximation in (1.7), and denoting by v_h the (time) discretized solution, we get

$$v_h(x,t) = h\Phi_g(x) + e^{\lambda h} v_h(x + h\Phi_f(x), t-h). \tag{2.3}$$

It is convenient to consider (2.3) only at discrete times $t_n = t_0 + nh$. Including the initial condition, we can rewrite (2.3) as

$$\begin{cases} v_h(x,t_n) = h\Phi_g(x) + e^{\lambda h} v_h(x + h\Phi_f(x), t_{n-1}), \\ v_h(x,t_0) = v_0(x). \end{cases} \tag{$\mathrm{EP_h}$}$$

Note that, for simplicity, the time step h has been kept constant, but the case of a variable time step requires only formal adaptations. Moreover, we will state the results for $t_0 = 0$. The convergence of v_h to v is proved by the following theorem (see [RM], [S], [FG]). Note that, here and in the sequel, the time step h is defined by $h \equiv (\bar t/n)$ for some $n \in \mathbb{N}$ and the vanishing of h is obtained for $n \to +\infty$.

THEOREM 2.1. *Assume* (h1), (h2). *Then, for any* n, ($\mathrm{EP_h}$) *has a unique solution* $v_h \in W^{1,\infty}(\mathbb{R}^N)$. *Moreover, as* $h \to 0$, $v_h(x,\bar t) \to v(x,\bar t)$ *uniformly on bounded sets of* $\mathbb{R}^N$.

We turn now to the estimation of the convergence rate for the error $\|v(\bar t) - v_h(\bar t)\|_\infty$. We will assume that

$$f \in C^p(\mathbb{R}^N), \tag{2.4}$$

$$g \in C^p(\mathbb{R}^N). \tag{2.5}$$

Moreover, we assume that the one-step scheme (2.2) is of order p, that is,

$$|y(h) - x - h\Phi_f(x)| \le Ch^{p+1}, \tag{2.6}$$

$$\left| \int_0^h e^{\lambda s} g(y(s))ds - h\Phi_g(x) \right| \le Ch^{p+1}. \tag{2.7}$$

Let us prove the convergence rate in $L^\infty(\Omega)$ for the scheme (EP_h).

THEOREM 2.2. *Assume that* (h1), (h2), (2.4)–(2.7) *hold, and that* $v_0 \in W^{1,\infty}(\mathbb{R}^N)$. *Then, for any* $\bar t \in [0,T]$, $\|v_h(\bar t) - v(\bar t)\|_\infty \le C\bar t h^p$.

Proof. We recall that under the previous assumptions, the solution v of (EP) is Lipschitz continuous on $\mathbb{R}^N \times [0,T]$. Comparing (1.7) and ($\mathrm{EP_h}$), and setting $\bar t = t_n = nh$, we have

$$\begin{cases} v(x,t_n) - v_h(x,t_n) = \displaystyle\int_0^h e^{\lambda s} g(y(s))ds - h\Phi_g(x) \\ \qquad\qquad + e^{\lambda h}\left[v(y(h),t_{n-1}) - v_h(x + h\Phi_f(x), t_{n-1})\right] \\ v(x,t_0) - v_h(x,t_0) = 0. \end{cases} \tag{2.8}$$

Recalling that $e^{\lambda h} \le 1$ and using (2.7), we can obtain from (2.8) the bound

$$\begin{aligned} &|v(x,t_n) - v_h(x,t_n)| \\ &\le |v(y(h),t_{n-1}) - v(x+h\Phi_f(x),t_{n-1})| + |v(x+h\Phi_f(x),t_{n-1}) \\ &\quad - v_h(x+h\Phi_f(x),t_{n-1})| + Ch^{p+1}, \end{aligned}$$

which gives, using the Lipschitz continuity of v and denoting by L_v its Lipschitz constant,

$$\|v(t_n) - v_h(t_n)\|_\infty \le L_v|y(h) - x - h\Phi_f(x)| + \|v(t_{n-1}) - v_h(t_{n-1})\|_\infty + Ch^{p+1}. \tag{2.9}$$

Using (2.6), and taking into account that $\|v(t_0) - v_h(t_0)\|_\infty = 0$, we get from (2.9)

$$\|v(t_n) - v_h(t_n)\|_\infty \le Cnh^{p+1},$$

which proves the theorem since $n = t_n/h$. ∎

Remark 1. When (2.6), (2.7) hold with different orders p', p'' it turns out from the proof of Theorem 2.2 that the contribution to the global error due to the one-step scheme is $O(h^{p'})$ and that due to the quadrature formula is $O(h^{p''})$. Then, the statement still holds true for $p = \min(p', p'')$.

It is important to note that the above proof relies on the regularity of the trajectories of (1.1) rather than on the regularity of the solution. This implies that a high rate of convergence can also be obtained for Lipschitz continuous solutions. This is a typical feature of these techniques.

3. Time discretization for the stationary equation and examples.

Throughout this section and section 6, we will make the standing assumption that $\lambda < 0$ in order to ensure the wellposedness of (SP). Applying the same ideas of the previous section in (1.7), we obtain the discretized equation:

$$v_h(x) = h\Phi_g(x) + e^{\lambda h} v_h(x + h\Phi_f(x)) \tag{$\mathrm{SP_h}$}$$

where v_h denotes the approximate solution. Note that ($\mathrm{SP_h}$) is in the form of a fixed-point equation.

We quote the following results, which are particular cases of the results obtained for the nonlinear case in [FF] (see also [BS], [C], and [CI]).

THEOREM 3.1. *Assume* (h1), (h2). *Then,* ($\mathrm{SP_h}$) *has a unique solution* $v_h \in W^{1,\infty}(\mathbb{R}^N)$. *Moreover, as* $h \to 0$, $v_h(x) \to v(x)$ *uniformly on bounded sets of* $\mathbb{R}^N$.

THEOREM 3.2. *Assume that* (h1), (h2), *and* (2.4)–(2.7) *hold, and that* $v \in W^{1,\infty}(\mathbb{R}^N)$. *Then, for* $0 < h < h_0$,

$$\|v - v_h\|_\infty \le Ch^p.$$

Before going on in our analysis, we give some examples of schemes in the form ($\mathrm{EP_h}$), ($\mathrm{SP_h}$).

Example 1 (Euler scheme). This is the simplest case. Adopting the notations already used, we set $\Phi_f(x) = f(x)$, $\Phi_g(x) = g(x)$. Then, ($\mathrm{EP_h}$), ($\mathrm{SP_h}$) read

$$\begin{cases} v_h(x, t_n) = hg(x) + e^{\lambda h} v_h(x + hf(x), t_{n-1}), \\ v_h(x, t_0) = v_0(x), \end{cases} \tag{3.1}$$

$$v_h(x) = hg(x) + e^{\lambda h} v_h(x + hf(x)). \tag{3.2}$$

Example 2 (Modified Euler scheme). For this scheme we have

$$\Phi_f(x, h) = f\left(x + \frac{h}{2} f(x)\right),$$

$$\Phi_g(x, h) = g\left(x + \frac{h}{2} f(x)\right) e^{\lambda h/2}.$$

The discrete equations become

$$\begin{cases} v_h(x, t_n) = hg\left(x + \frac{h}{2} f(x)\right) e^{\lambda h/2} + e^{\lambda h} v_h\left(x + hf\left(x + \frac{h}{2} f(x)\right), t_{n-1}\right), \\ v_h(x, t_0) = v_0(x), \end{cases} \tag{3.3}$$

$$v_h(x) = hg\left(x + \frac{h}{2} f(x)\right) e^{\lambda h/2} + e^{\lambda h} v_h\left(x + hf\left(x + \frac{h}{2} f(x)\right)\right). \tag{3.4}$$

Example 3 (Heun's formula). We can set here

$$\Phi_f(x, h) = \frac{1}{2} f(x) + \frac{1}{2} f(x + hf(x)),$$

$$\Phi_g(x, h) = \frac{1}{2} g(x) + \frac{1}{2} g(x + hf(x)) e^{\lambda h}.$$

We obtain the discrete equations

$$\begin{cases} v_h(x, t_n) = \frac{h}{2} g(x) + \frac{h}{2} g(x + hf(x)) e^{\lambda h} + e^{\lambda h} v_h\left(x + \frac{h}{2} f(x) + \frac{h}{2} f(x + hf(x)), t_{n-1}\right), \\ v_h(x, t_0) = v_0(x), \end{cases} \tag{3.5}$$

$$v_h(x) = \frac{h}{2} g(x) + \frac{h}{2} e^{\lambda h} g(x + hf(x)) + e^{\lambda h} v_h\left(x + \frac{h}{2} f(x) + \frac{h}{2} f(x + hf(x))\right). \tag{3.6}$$

Example 4 (Fourth order Runge–Kutta scheme). In this case we set

$$K_0 = f(x),$$

$$K_1 = f\left(x + h\frac{K_0}{2}\right),$$

$$K_2 = f\left(x + h\frac{K_1}{2}\right),$$

$$K_3 = f(x + hK_2),$$

$$\Phi_f(x,h) = \frac{1}{6}(K_0 + 2K_1 + 2K_2 + K_3),$$

$$\Phi_g(x,h) = \frac{1}{6}\left(g(x) + 2e^{\lambda h/2} g\left(x + h\frac{K_0}{2}\right) + 2e^{\lambda h/2} g\left(x + h\frac{K_1}{2}\right) + e^{\lambda h} g(x + hK_2)\right).$$

Then, the discretized equations are obtained using the above expressions of Φ_f, Φ_g into $(\mathrm{EP_h})$, $(\mathrm{SP_h})$.

4. Space discretization: General framework and evolution problem. In the previous sections we have shown how to obtain an approximation v_h of the solution v in $W^{1,\infty}(\mathbb{R}^N)$, by a time discretization of the evolution problem. This gives $(\mathrm{EP_h})$ in the form of a forward difference equation and $(\mathrm{SP_h})$ in the form of a fixed-point equation. For the construction of the numerical algorithm we need to compute the solution in a bounded domain Ω instead of $\mathbb{R}^N$. Without giving details, we recall that the wellposedness of (EP), (SP) on a bounded set requires either that this set is invariant for the flow of the vector field f, that is,

$$f(x) \cdot \nu(x) < c < 0 \tag{4.1}$$

for any $x \in \partial\Omega$ (here, $\nu(x)$ denotes the outward normal), or that a boundary condition is imposed at all points $x \in \partial\Omega$ not satisfying (4.1) (see section 7).

The space discretization of v_h is obtained by constructing a grid on Ω with space step k (not necessarily constant) and setting

$$v_h^k(x, t_n) = \sum_{j \in Q} v_j(t_n)\psi_j(x) = \sum_{j \in Q} v_j^n \psi_j(x) \tag{4.2a}$$

for the evolution problem, and

$$v_h^k(x) = \sum_{j \in Q} v_j \psi_j(x) \tag{4.2b}$$

for the stationary problem, where v_h^k is the space discretization of v_h, $Q = \{1, \ldots, q\}$ is the set of indices of the grid nodes x_j, and $\{\psi_j(x)\}_{j \in Q}$ is a basis of bounded functions such that, for any $j \in Q$

$$\psi_j(x_j) = 1\ , \quad \psi_j(x_m) = 0 \quad (m \neq j). \tag{4.3a}$$

We will also define the symmetric (mass) matrix $M(\psi)$ with entries

$$m_{ij} = \int_\Omega \psi_i(x)\psi_j(x)dx, \qquad (i, j = 1, \ldots, q).$$

Using the basis $\{\psi_i\}$ we can define a projection operator P_k as

$$P_k w(x,t) := \sum_{m \in Q} w(x_m, t)\psi_m(x) \tag{4.3b}$$

for time-dependent functions. Similarly, for time-independent functions we define

$$P_k w(x) := \sum_{m\in Q} w(x_m)\psi_m(x).$$

Note that the definition of the projection in terms of a linear combination of the basis function is only one among several possibilities, more general choices are possible (e.g., monotone interpolations). We will assume that, for any $w \in W^{1,\infty}(\Omega)$,

$$\lim_{k\to 0} \|w - P_k w\|_\infty = 0. \tag{4.3c}$$

Let us now give some examples of bases for space discretization. For simplicity, we will refer to the case of $\mathbb{R}^1$.

Example 1 (Finite elements basis). This case does not require a structured grid. A first choice of the basis functions could be

$$\psi_j(x) = \begin{cases} 0 & \text{if} \quad x < x_j - k/2, \\ 1 & \text{if} \quad x_j - k/2 < x < x_j + k/2, \\ 0 & \text{if} \quad x > x_j + k/2, \end{cases}$$

namely, piecewise constant functions. A more efficient choice is

$$\psi_j(x) = \begin{cases} 0 & \text{if} \quad x < x_{j-1}, \\ 1/k(x - x_{j-1}) & \text{if} \quad x_{j-1} < x < x_j, \\ -1/k(x - x_{j+1}) & \text{if} \quad x_j < x < x_{j+1}, \\ 0 & \text{if} \quad x > x_{j+1}, \end{cases}$$

that is, P_1 (piecewise linear) finite elements. Following the same principle, one could consider piecewise polynomial functions. This requires increasing the number of nodes (i.e., the number of degrees of freedom) within a space step.

Example 2 (Polynomial basis). It is also possible to consider basis functions in the form of Lagrange polynomials on the nodes x_j, namely,

$$\psi_j(x) = \frac{(x - x_1)\cdots(x - x_{j-1})(x - x_{j+1})\cdots(x - x_q)}{(x_j - x_1)\cdots(x_j - x_{j-1})(x_j - x_{j+1})\cdots(x_j - x_q)}.$$

This definition satisfies (4.3), but it will not satisfy the bound

$$\|\psi_j\|_\infty = 1 \tag{4.4}$$

unless the nodes x_j are chosen as the zeroes of a Legendre polynomial of degree q on the interval Ω. In this case the space step k is of the same order of magnitude as $1/q$.

If (4.1) holds, by the consistency assumption (h2a) it follows that for h small enough

$$\Phi_f(x) \cdot \nu(x) < 0 \tag{4.5}$$

for any $x \in \partial\Omega$.

In the sequel $|W|_2 := (\sum_i w_i^2)^{1/2}$ will denote the usual 2-norm for a vector $W \in \mathbb{R}^q$ and $|W|_\infty = \max_i |w_i|$. The corresponding matrix norms will be denoted by

$$|A|_\infty = \max_i \sum_j |a_{ij}|, \quad |A|_2 = \sqrt{\rho(A^T A)}$$

where $\rho(B)$ denotes the spectral radius of the matrix B. We will also define

$$V := \begin{pmatrix} v_1 \\ \vdots \\ v_q \end{pmatrix} \ , \quad V^n := \begin{pmatrix} v_1^n \\ \vdots \\ v_q^n \end{pmatrix} \ , \quad G := \begin{pmatrix} h\Phi_g(x_1) \\ \vdots \\ h\Phi_g(x_q) \end{pmatrix},$$

$$\Psi := (\psi_{ij}) = \begin{pmatrix} \psi_1(x_1 + h\Phi_f(x_1)) & \dots & \psi_q(x_1 + h\Phi_f(x_1)) \\ \vdots & & \vdots \\ \psi_1(x_q + h\Phi_f(x_q)) & \dots & \psi_q(x_q + h\Phi_f(x_q)) \end{pmatrix}.$$

We turn now to the evolution equation (EP). Using the approximation (4.2a) and assuming (4.1)–(4.3), we can obtain a fully discrete version of (EP) by collocating ($\mathrm{EP_h}$) in the nodes x_j, that is,

$$(\mathrm{EP_h^k}) \qquad \begin{cases} v_j^n = h\Phi_g(x_j) + e^{\lambda h} \displaystyle\sum_{m \in Q} v_m^{n-1} \psi_m(x_j + h\Phi_f(x_j)) \ , \quad j \in Q, \\ v_j^0 = v_0(x_j), \end{cases}$$

or, more briefly,

$$V^n = G + e^{\lambda h} \Psi V^{n-1} \tag{4.6}$$

with a proper initial condition V^0. Note that the form (4.6) is a direct consequence of our choice of the projection operator and requires only a matrix–vector product for each time step. This form allows an easy parallel implementation.

Remark 2 (Monotonicity). By applying (4.6) with two different initial vectors V^0 and W^0 such that $v_j^0 > w_j^0$ for any j, we obtain

$$V^1 - W^1 = e^{\lambda h} \Psi (V^0 - W^0).$$

Then, the monotonicity of the scheme requires Ψ to have positive elements; this requires in turn that functions ψ must be positive. In the previous examples, this condition holds for P_0 or P_1 finite elements, but not for higher-order finite elements or for Lagrange polynomials. However, one can also adopt a nonlinear interpolation which preserves monotonicity as in [RW] (see section 5). This requires some modifications to the present theory which will be considered in a future paper.

Remark 3. If $|\Psi|_2 \le 1$ then the sequence V^n has nonincreasing 2-norm for any n. If $|\Psi|_\infty \le 1$, then both norms are nonincreasing. We will assume for the moment that at least one of these conditions is satisfied, so that the scheme is stable (respectively, in L^∞ and L^2) and refer to section 5 for stability analysis. On the other hand, it is easy to prove that, under basic assumptions (and, in particular, (h2a,b) and (4.3c)), ($\mathrm{EP_h^k}$) is also consistent, and, therefore, the classical theory of finite differences (see [RM]) applies.

Our interest, however, is to obtain the convergence rate of v_h^k in L^∞ and in L^2. In order to make clear the dependence on the initial position, we will denote by $y_x(\cdot) = y(x, \cdot)$ the solution trajectory of (1.1) and we will identify $y_j(\cdot) = y(x_j, \cdot)$.

THEOREM 4.1. *Assume* (h1), (h2), (2.4)–(2.7), (4.1), *and* (4.3). *Assume, moreover, that* $v_0 \in W^{1,\infty}(\Omega)$ *and that* $\|v(t) - P_k v(t)\|_\infty \le E(k)$ *for* $t \in [0, \bar{t}]$. *The following estimates hold true for some positive constant* C:

(i) *If* $\sum_{j \in Q} |\psi_j(x)| \le 1$, *then for any* $h < h_0$

$$\left\| v_h^k(\bar{t}) - v(\bar{t}) \right\|_\infty \le C \left(h^p + \frac{1}{h} E(k) \right). \tag{4.7}$$

(ii) *Let* $|\Psi|_2 \le 1$ *and* μ_2 *be a positive constant such that* $|M(\psi)|_2 \le \mu_2 k^N$ *for any* k. *Then, for any* $h < h_0$,

$$\|v_h^k(\bar t) - v(\bar t)\|_2 \le C\left(h^p + \frac{1}{h}E(k)\right). \tag{4.8}$$

Proof. (i) Note that $\sum_{j\in Q} |\psi_j(x)| \le 1$ implies $|\Psi|_\infty \le 1$. By the triangular inequality we have

$$\begin{aligned}\left\|v_h^k(t_n) - v(t_n)\right\|_\infty &\le \left\|v_h^k(t_n) - P_k v(t_n)\right\|_\infty + \|P_k v(t_n) - v(t_n)\|_\infty \\ &\le \left\|\sum_{m\in Q}(v_m^n - v(x_m,t_n))\psi_m\right\|_\infty + E(k) \\ &\le |V^n - W^n|_\infty + E(k)\end{aligned} \tag{4.9}$$

with V^n defined by (4.6), and W^n having in its mth component $v(x_m, t_n)$. By (1.7) and ($\mathrm{EP}_\mathrm{h}^\mathrm{k}$), we obtain

$$\begin{aligned}|v_j^n - v(x_j,t_n)| &\le \left|h\Phi_g(x_j) - \int_0^h e^{\lambda s} g(y_j(s))ds\right| \\ &\quad + e^{\lambda h}\left|\sum_{m\in Q} v_m^{n-1}\psi_m(z_j) - v(y_j(h), t_{n-1})\right|\end{aligned} \tag{4.10}$$

(here and in the sequel $z_j = x_j + h\Phi_f(x_j)$). Hence,

$$\begin{aligned}|v_j^n - v(x_j,t_n)| &\le Ch^{p+1} + e^{\lambda h}\left|\sum_{m\in Q}(v_m^{n-1} - v(x_m,t_{n-1}))\psi_m(z_j)\right| \\ &\quad + e^{\lambda h}\left|P_k v(z_j,t_{n-1}) - v(z_j,t_{n-1})\right| \\ &\quad + e^{\lambda h}\left|v(z_j,t_{n-1}) - v(y_j(h),t_{n-1})\right| \\ &\le C(1+e^{\lambda h}L_v)h^{p+1} + e^{\lambda h}\left|\sum_{m\in Q}(v_m^{n-1} - v(x_m,t_{n-1}))\psi_m(z_j)\right| + e^{\lambda h}E(k).\end{aligned} \tag{4.11}$$

Therefore, under our basic assumptions, we have

$$|V^n - W^n|_\infty \le C\, h^{p+1} + e^{\lambda h}E(k) + e^{\lambda h}|\Psi|_\infty |V^{n-1} - W^{n-1}|_\infty.$$

Iterating back, and taking into account that $|\Psi|_\infty \le 1$ and that $V^0 = W^0$, we obtain

$$\begin{aligned}|V^n - W^n|_\infty &\le Ch^{p+1} + e^{\lambda h}E(k) + e^{\lambda h}|V^{n-1} - W^{n-1}|_\infty \\ &\le \cdots \le \\ &(1 + e^{\lambda h} + \cdots + e^{(n-1)\lambda h})(Ch^{p+1} + e^{\lambda h}E(k)).\end{aligned}$$

If $\lambda > 0$, then the sum appearing in the right-hand side may be bounded by

$$(1 + e^{\lambda h} + \cdots + e^{(n-1)\lambda h}) = \frac{e^{n\lambda h} - 1}{e^{\lambda h} - 1} \le \frac{e^{n\lambda h}}{\lambda h}.$$

If $\lambda = 0$, then $(1 + e^{\lambda h} + \cdots + e^{(n-1)\lambda h}) = n$. If $\lambda < 0$, then

$$(1 + e^{\lambda h} + \cdots + e^{(n-1)\lambda h}) \le \frac{1}{1 - e^{\lambda h}} = \frac{1}{\lambda h} + o(1/h).$$

This completes the proof of (4.7) since $\bar{t} = nh$.

(ii) The same argument used in i) leads to

$$\begin{aligned} \|v_h^k(t_n) - v(t_n)\|_2 &\le \left(\int_\Omega \left(\sum_{m\in Q} (v_m^n - v(x_m, t_n))\psi_m(x) \right)^2 dx \right)^{1/2} + C_\Omega E(k) \\ &\le ((V^n - W^n)^T M(\psi)(V^n - W^n))^{1/2} + C_\Omega E(k) \end{aligned}$$

where $C_\Omega \equiv meas(\Omega)^{1/2}$. By the assumption on $M(\psi)$ we then get

$$\|v_h^k(t_n) - v(t_n)\|_2 \le \sqrt{\mu_2} k^{N/2} |V^n - W^n|_2 + C_\Omega E(k).$$

By (4.10), we can obtain the bound

$$|V^n - W^n|_2 \le k^{-N/2}(C\, h^{p+1} + e^{\lambda h} E(k)) + e^{\lambda h} |\Psi|_2 |V^{n-1} - W^{n-1}|_2.$$

As in (i), iterating back and taking into account that $|\Psi|_2 \le 1$ and that $V^0 = W^0$, we obtain

$$|V^n - W^n|_2 \le k^{-N/2}(1 + e^{\lambda h} + \cdots + e^{(n-1)\lambda h})(Ch^{p+1} + e^{\lambda h} E(k)).$$

Then, the same analysis yields

$$\|v_h^k(t_n) - v(t_n)\|_2 \le \sqrt{\mu_2} C \left(h^p + \frac{1}{h} E(k) \right). \qquad \square$$

As far as the L^∞ rate of convergence is concerned, the previous theorem is an extension of the result in [FG] related to first-order schemes.

Remark 4. Let f and g depend also on t, that is $f = f(x,t)$, $g = g(x,t)$. This is the case, e.g., when dealing with a more general attenuation term in (EP) like $\lambda(v(x,t) - \tilde{v}(x,t))$ ($\tilde{v}$ here is a known function). The above analysis may still be applied (with obvious formal changes) and gives similar results provided the dependence on t is smooth enough to preserve the order in (2.6), (2.7). Clearly, the matrix Ψ in (4.6) is no longer constant.

Remark 5. In applications (e.g., when dealing with experimental data or using (EP), (SP) as a linearized step for solving a nonlinear equation), f and g are usually obtained from an interpolation. Then, the function Φ to be used in (2.2) actually refers to the interpolated functions

$$P_k f(x) := \sum_{m\in Q} f(x_m)\psi_m(x) \ , \quad P_k g(x) := \sum_{m\in Q} g(x_m)\psi_m(x)$$

(we have assumed here that f and g are interpolated in the same basis of v), so that in this case (2.6) implies

$$|y(h) - x - h\Phi_{P_k f}(x)| \le |y(h) - x - h\Phi_f(x)| + h|\Phi_f(x) - \Phi_{P_k f}(x)| \le C(h^{p+1} + hE(k)).$$

It is apparent that using this bound instead of (2.6) or (2.7) does not change the estimates (4.7) and (4.8), since the term $E(k)$ is asymptotically negligible with respect to $E(k)/h$. If f and g are interpolated on a different basis, then a further error term should be taken into consideration.

Let us now denote by $B(x, R)$ the ball of R^N with radius R, centered at x, and define the "cylindrical" set

$$\omega(x, t, \rho) := \{(z, s) \in R^{N+1} : s \in [0, t],\ z \in B(y(x, s), \rho)\}.$$

A widely observed feature of semi-Lagrangian schemes (see [SC]) is the fact that errors are usually localized in a small neighborhood of discontinuities. The following theorem improves Theorem 4.1 in this respect by estimating the (local) convergence rate of v_h^k at a point x.

THEOREM 4.2. *Assume* (h1), (h2), (2.4)–(2.7), (4.1), *and* (4.3). *Assume, moreover, that $v \in W^{1,\infty}(\omega(x, \bar{t}, \rho))$ for some $\rho > 0$. Let $|\Psi|_\infty \leq 1$,* supp $\psi_j \subset B(x_j, k)$, *and*

$$\sup_{\omega(x,\bar{t},\rho)} |v(z, t) - P_k v(z, t)| \leq E(x, \bar{t}, \rho, k);$$

then, for any h, k such that $h < h_0$, $k = o(h)$, the following estimate holds:

$$|v_h^k(x, \bar{t}) - v(x, \bar{t})| \leq C(\bar{t}) \left(h^p + \frac{1}{h} E(x, \bar{t}, \rho, k) \right), \tag{4.12}$$

where $C(s) = C$ if $\lambda < 0$, $C(s) = Cs$ if $\lambda = 0$, $C(s) = Ce^{\lambda s}$ if $\lambda > 0$.

Proof. For simplicity, we will give the proof for the case $\lambda \leq 0$.

Operating as in the second case of the previous proof, we obtain

$$\begin{aligned} |v_h^k(x, t_n) - v(x, t_n)| &\leq |v_h^k(x, t_n) - P_k v(x, t_n)| + |P_k v(x, t_n) - v(x, t_n)| \\ &\leq \left| \sum_{j \in Q(x,k)} (v_j^n - v(x_j, t_n)) \psi_j(x) \right| + E(x, \bar{t}, \rho, k) \\ &\leq \max_{j \in Q(x,k)} |v_j^n - v(x_j, t_n)| + E(x, \bar{t}, \rho, k) \end{aligned} \tag{4.13}$$

where $Q(x, k)$ denotes the set of indices j such that $x_j \in B(x, k)$ (note that all other terms in (4.13) may be omitted since the supports of the corresponding functions ψ_j do not contain x). Our aim is to prove that the error at $(x, \bar{t})$ depends only on the behavior of the solution in the set $\omega(x, \bar{t}, \rho)$. To this end, we assume now that x_j is a node such that $x_j \in B(y((n - s)h), R_s)$ for a generic R_s ($s = 1, \ldots, n$). The plan is to estimate the error at (x_j, t_s) as a function of the errors at (x_m, t_{s-1}), with $x_m \in B(y((n - s + 1)h), R_{s-1})$ and R_{s-1} to be determined. By (1.7) and $(\mathrm{EP}_\mathrm{h}^\mathrm{k})$ we obtain

$$\begin{aligned} |v_j^s - v(x_j, t_s)| \leq &\left| h\Phi_g(x_j) - \int_0^h e^{\lambda s} g(y_j(s)) ds \right| \\ &+ e^{\lambda h} \left| \sum_{m \in Q(z_j,k)} v_m^{s-1} \psi_m(z_j) - v(y_j(h), t_{s-1}) \right|. \end{aligned} \tag{4.14}$$

Hence, using the triangular inequality,

$$\begin{aligned} |v_j^s - v(x_j, t_s)| \leq Ch^{p+1} &+ \left| \sum_{m \in Q(z_j,k)} (v_m^{s-1} - v(x_m, t_{s-1})) \psi_m(z_j) \right| \\ &+ |P_k v(z_j, t_{s-1}) - v(z_j, t_{s-1})| \\ &+ |v(z_j, t_{s-1}) - v(y_j(h), t_{s-1})|. \end{aligned} \tag{4.15}$$

Taking into account that $|\Psi|_\infty \le 1$, we obtain from (4.15)

$$|v_j^s - v(x_j,t_s)| \le Ch^{p+1} + \max_{m\in Q(z_j,k)} \left|v_m^{s-1} - v(x_m,t_{s-1})\right| + \sup_{z\in B(y((n-s+1)h),\bar R)} |P_k v(z,t_{s-1}) - v(z,t_{s-1})| + Ch^{p+1} \sup_{z\in B(y((n-s+1)h),\bar R)} |\nabla v(z,t_{s-1})|. \tag{4.16}$$

Here, we have set $\bar R := e^{L_f h}R_s + Ch^{p+1}$. Note that by Gronwall's lemma $|y_1(h) - y_2(h)| \le e^{L_f h}|x_1 - x_2|$, and, therefore, all the points z_j, $y_j(h)$ belong to the ball $B(y((n-s+1)h), \bar R)$. To complete the step, we only need to define

$$R_{s-1} := e^{L_f h}R_s + Ch^{p+1} + k \tag{4.17}$$

and rewrite (4.16) in the form

$$|v_j^s - v(x_j,t_s)| \le C(1+\bar L_v)h^{p+1} + \max_{m\in Q(y((n-s+1)h),R_{s-1})} \left|v_m^{s-1} - v(x_m,t_{s-1})\right| + \sup_{z\in B(y((n-s+1)h),R_{s-1})} |P_k v(z,t_{s-1}) - v(z,t_{s-1})| \tag{4.18}$$

(here, $\bar L_v$ denotes the Lipschitz constant of $v(t_{s-1})$ in the ball $B(y((n-s+1)h), R_{s-1})$).

To iterate back the estimation, note that $R_s < R_{s-1}$ and that by (4.13), $R_n = k$. Using (4.17) and recalling that $n = \bar t/h$ we have, for any $s = 1,\dots,n$:

$$R_s < R_0 = e^{L_f \bar t}k + (1 + e^{L_f h} + \cdots + e^{(n-1)L_f h})(Ch^{p+1} + k)$$

$$< e^{L_f \bar t}k + \frac{e^{L_f \bar t}}{L_f}\left(Ch^p + \frac{k}{h}\right)$$

so that $h, k, k/h \to 0$ imply that $R_0 \to 0$ and, in particular, that $R_0 < \rho$ for h small enough.

We obtain then, iterating back (4.18) with $s = n$, and denoting by L_v the Lipschitz constant of v in the set $\omega(x,\bar t,\rho)$

$$|v_j^n - v(x_j,t_n)| \le C(1+L_v)h^{p+1} + \max_{m\in Q(y_j(h),R_{n-1})} |v_m^{n-1} - v(x_m,t_{n-1})| + E(x,\bar t,\rho,k)$$

$$\le 2C(1+L_v)h^{p+1} + \max_{m\in Q(y_j(2h),R_{n-2})} |v_m^{n-2} - v(x_m,t_{n-2})| + 2E(x,\bar t,\rho,k)$$

$$\le \cdots \le C(1+L_v)nh^{p+1} + nE(x,\bar t,\rho,k) = C(1+L_v)\bar t h^p + \frac{\bar t}{h}E(x,\bar t,\rho,k)$$

which proves (4.12). ☐

Remark 6. The typical application of Theorem 4.2 is to provide the rate of convergence of the approximation in regions where the exact solution is regular. If f and g are smooth enough (say, $f, g \in C^\infty$), then the regularity of the solution is preserved along the characteristics. Thus, if x is internal to the region of smoothness at time $\bar t$, then it is possible to find a radius ρ such that $v(x,t)$ is smooth in the whole set $\omega(x,\bar t,\rho)$. This implies that the "local" interpolation error $E(x,\bar t,\rho,k)$ (related to the error at $(x,\bar t)$) may decrease faster than the "global" interpolation error $E(k)$ used in Theorem 4.1 (and related to the L^∞ error). Note that this result requires that the functions ψ_i have compact support, and, therefore, apply only to finite elements-type discretizations.

5. Stability. This section is devoted to the stability analysis of the schemes. We will also examine some extensions of the previous theory which may improve stability. Although we are not presently able to give an ultimate proof of stability for the most general setting, the following analysis covers a number of interesting cases.

5.1. L^∞ stability. A first (well-known) case which shows L^∞ stability is the case of low-order interpolations. More precisely, it is easy to check that for finite element interpolations of type P_0, P_1, Q_1 (piecewise linear or bilinear) the matrix Ψ satisfies the bound $|\Psi|_\infty = 1$ since the functions ψ are positive and have unitary sum.

A second situation which presents L^∞ stability is the case in which high-order monotone reconstructions are used to interpolate in the upstream point. In this case, the reconstructed approximate solution has no longer the linear form (4.2), and the scheme ($\mathrm{EP}_\mathrm{h}^\mathrm{k}$) (or ($\mathrm{SP}_\mathrm{h}^\mathrm{k}$)) itself becomes nonlinear. We rewrite this nonlinear version in the compact form of (4.6), namely,

$$V^n = G + e^{\lambda h}\Psi(V^{n-1}) \tag{5.1}$$

with a nonlinear function $\Psi : \mathbb{R}^q \to \mathbb{R}^q$. Although stability is ensured by the monotonicity of the interpolation, we cannot apply the classical Lax–Richtmeyer theorem to prove convergence since the scheme is nonlinear. However, we can consider the set of all solutions obtained by means of piecewise constant interpolations satisfying (4.3). All such solutions (as the previous analysis shows) are convergent to v, and, therefore, their supremum $\overline{V}$ and infimum $\underline{V}$ also converge to v. Since the solution V of (5.1) satisfies

$$\underline{V}^n \le V^n \le \overline{V}^n,$$

we obtain the uniform convergence of V^n to $v(t_n)$. By setting up a more formal proof, one could prove that the convergence estimate is given by (4.7) with $E(k) = k$. This estimate could also be improved by a sharper construction of the sub and supersolutions $\underline{V}$ and $\overline{V}$; we must note, however, that monotone interpolations cannot be more than second-order accurate in presence of local extrema.

In the general linear high-order case, unconditional L^∞ stability with respect to the time step h (but not to the space step k) can be proved as follows.

THEOREM 5.1. *Assume k is fixed, and* (4.3) *holds. Then the scheme* ($\mathrm{EP}_\mathrm{h}^\mathrm{k}$) *is stable for any $h > 0$.*

Proof. For $h \to 0$, the elements ψ_{ij} of the matrix Ψ may be written as

$$\psi_{ij} = \psi_j(x_i + h\Phi_f(x_i)) = \delta_{ij} + h a_{ij} \cdot f(x_i) + o(h) \tag{5.2}$$

where δ_{ij} is the Kroenecker symbol, $a_{ij} = \nabla\psi_j(x_i)$ if ψ_j is smooth in x_i and is a generalized gradient otherwise.

Therefore,

$$|\Psi|_\infty = \max_i \sum_j |\psi_{ij}| \le \max_i \Big[1 + h\sum_j |a_{ij}\cdot f(x_i)| + o(h)\Big] \le 1 + Ch \tag{5.3}$$

and the statement follows from a classical stability result for finite difference schemes (see [RM]). It could also be noted that, setting $h = T/n$ and using (5.3), we may give the bound

$$|\Psi^n|_\infty \le \left(1 + \frac{CT}{n}\right)^n \le e^{CT}. \qquad \square \tag{5.4}$$

Remark 7. Unfortunately, the stability bound (5.4) is useless for proving convergence estimates since the constant C has a critical dependence on k (for example, $C = \bar{C}/k$ for finite elements interpolations, even on regular grids).

5.2. L^2 stability. The classical tool for proving L^2 stability is Fourier analysis. As it is known, this technique is rigorous only for the case of constant coefficient equations in $\mathbb{R}^1$, with evenly spaced nodes and periodic conditions. Nonetheless, it is usually considered a significant indication for more general cases as well (we refer to [VB], [RM] for the general theory). Elementary solutions are written in the form

$$v_j^n = (\rho_j)^n e^{\boldsymbol{i}\omega jk} \tag{5.5}$$

where $\boldsymbol{i}$ is the imaginary unit and ρ_j a complex number (amplification factor). For (5.5) to be a stable solution of $(\mathrm{EP}_\mathrm{h}^\mathrm{k})$, it is required that (5.5) satisfies $(\mathrm{EP}_\mathrm{h}^\mathrm{k})$ with $|\rho_j| \le 1$ for any j and ω (Von Neumann stability condition).

Rewriting (5.5) for v_j^{n+1} we get

$$v_j^{n+1} = (\rho_j)^{n+1} e^{\boldsymbol{i}\omega jk}. \tag{5.6}$$

Then, using (4.6) with $\lambda = 0$, $G = 0$ (this is not restrictive for stability analysis), by (5.5), (5.6) we obtain

$$(\rho_j)^{n+1} e^{\boldsymbol{i}\omega jk} = (\rho_j)^n \sum_m \psi_{jm} e^{\boldsymbol{i}\omega mk},$$

that is,

$$\rho_j = \sum_m \psi_{jm} e^{\boldsymbol{i}\omega(m-j)k}. \tag{5.7}$$

Therefore, the Von Neumann stability condition $|\rho_j| \le 1$ reads

$$\left[\Big(\sum_m \psi_{jm} \cos(\omega(m-j)k) \Big)^2 + \Big(\sum_m \psi_{jm} \sin(\omega(m-j)k) \Big)^2 \right]^{1/2} \le 1. \tag{5.8}$$

We recall that for classical difference schemes where Fourier stability analysis applies,

$$\psi_{jm} = \psi_{j+1,m+1}. \tag{5.9}$$

For schemes which do not satisfy (5.9), as well as for nonconstant coefficient equations, this kind of analysis becomes less rigorous. A possible approach in this case is to prove conditions such that (5.8) would hold for any index j (obtaining the so-called *local stability analysis*).

In our case, defining $z_j := x_j + h\Phi_f(x_j)$, we have

$$\psi_{jm} = \psi_m(z_j).$$

For locally polynomial interpolations of degree r, we can make (5.7) more explicit by setting $k = 1$, and the basis functions ψ_m defined as the $r+1$ Lagrange polynomials built on the nodes $0, 1, \ldots, r$, that is,

$$\psi_{jm} = \frac{z_j(z_j-1)\cdots(z_j-m-1)(z_j-m+1)\cdots(z_j-r)}{m(m-1)\cdots 2\cdot 1(-1)\cdots(m-r)}. \tag{5.10}$$

It is very unlikely that (5.8), (5.10) could be solved in an explicit form. We can, nevertheless, plot the left-hand side of (5.8) on the set $[0,r]\times[0,2\pi]$ of the $z_j-\omega$ plane, looking for regions in which (5.8) could be satisfied uniformly with respect to ω. The result is plotted in Figure 1 for $r=1,2,3,4$ along with the level curve $\rho=1$.

With interpolations of higher order, the behavior of ρ is similar: the amplification factor satisfies the Von Neumann condition in the strip $[r/2-1,r/2+1]\times[0,2\pi]$ if r is even, and in the strip $[s,s+1]\times[0,2\pi]$ (s being the integer part of $r/2$) if r is odd. The Von Neumann condition is not satisfied elsewhere.

Therefore, interpolations of order 1 or 2 are L^2 stable. Interpolations of order 3 and higher are stable only provided $z_j\in[r/2-1,r/2+1]$ if r is even, or $z_j\in[s,s+1]$ if r is odd. Since z_j depends on f and h, this condition cannot be ensured for any choice of h and k, at least if the interpolation is made in a fixed basis as assumed in the previous sections. However, this suggests a different way of interpolating that would lead to an L^2 stable scheme. For example, this could be done for $r=3$ by using the four nodes (two on the left and two on the right) around z_j, that is (defining l as the index of the node just on the left of z_j),

(5.11)

$$\psi_{j,l-1}=\frac{(z_j-x_l)(z_j-x_{l+1})(z_j-x_{l+2})}{(x_{l-1}-x_l)(x_{l-1}-x_{l+1})(x_{l-1}-x_{l+2})}=-\frac{(z_j-x_l)(z_j-x_{l+1})(z_j-x_{l+2})}{6k^3}$$

$$\vdots \qquad\qquad\qquad\qquad \vdots$$

$$\psi_{j,l+2}=\frac{(z_j-x_{l-1})(z_j-x_l)(z_j-x_{l+1})}{(x_{l+2}-x_{l-1})(x_{l+2}-x_l)(x_{l+2}-x_{l+1})}=\frac{(z_j-x_{l-1})(z_j-x_l)(z_j-x_{l+1})}{6k^3}.$$

We point out that this way of defining the matrix Ψ satisfies (5.9) for constant coefficients equations and is, in fact, the usual choice in the semi-Lagrangian literature.

Remark 8. We have implicitly assumed that the stability analysis does not change if the set of coefficients ψ_{jm} is replaced by a set $\psi_{j,m+d}$ with a shifted column index, corresponding to a nonneighboring cell of the space grid. In fact, in this case, (5.7) reads

$$\rho_j=\sum_m\psi_{j,m+d}\;e^{i\omega(m-j)k}=e^{-i\omega dk}\sum_m\psi_{j,m+d}\;e^{i\omega(m+d-j)k},$$

and the result is to multiply the right-hand side of (5.7) by the pure phase term $e^{-i\omega dk}$. This is a crucial remark to show that stability is not affected by large Courant numbers that cause the difference $z_j-x_j=h\Phi_f(x_j)$ to be much larger then the space step.

6. Space discretization: The stationary problem. Let us consider the discretized stationary equation ($\mathrm{SP_h}$). As we said in section 1 in the analysis of the stationary problem we will always assume $\lambda<0$. Applying (4.2b), we obtain the (fully discrete) equation:

$$(\mathrm{SP_h^k})\qquad v_j=h\Phi_g(x_j)+e^{\lambda h}\sum_{m\in Q}v_m\psi_m(x_j+h\Phi_f(x_j))\ ,\quad j\in Q,$$

which amounts to a linear system of equations in the unknowns v_j which may be rewritten as

$$(6.1)\qquad V=T_h^k(V):=G+e^{\lambda h}\Psi V.$$

Thus, the fully discrete approximation for the stationary equation is again the asymptotic solution for the corresponding approximation of the evolution equation. We prove the following theorem.

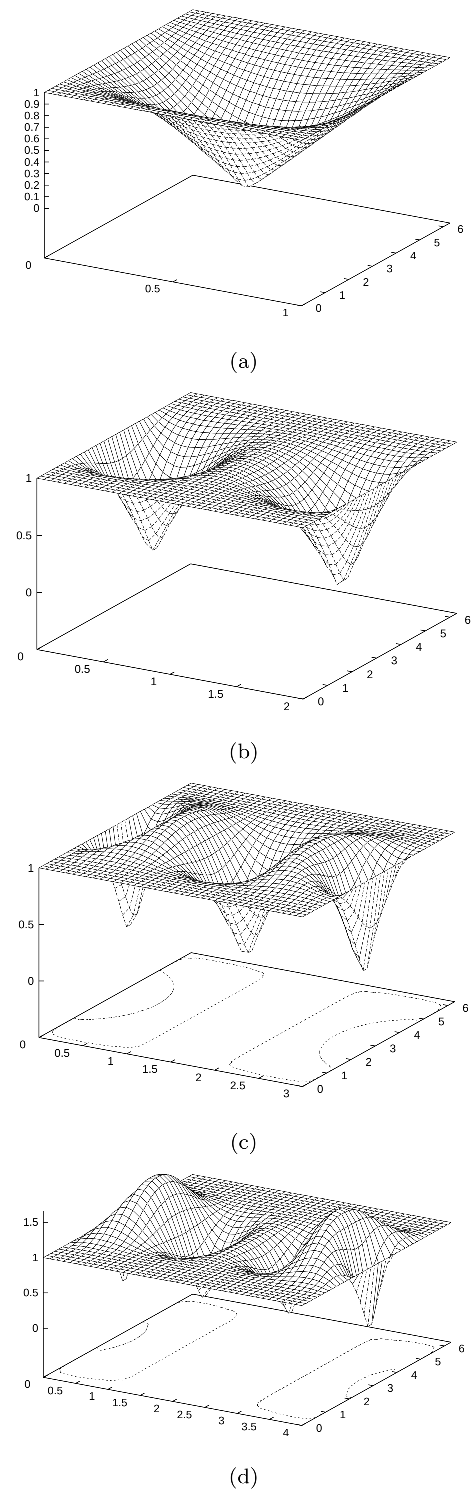


FIG. 1. *Amplification factors for interpolation:* (a) *order* 1, (b) *order* 2, (c) *order* 3, *and* (d) *order* 4.

THEOREM 6.1. *Let* (h1), (h2), (4.1), *and* (4.3) *hold true. Moreover, assume that*

$$|\Psi|_p \le 1, \qquad \textit{for } p = 2, \infty.$$

Then, $(\mathrm{SP_h^k})$ *has a unique solution.*

Proof. It is a simple calculation to show that the right-hand side of (6.1) is a contraction mapping with respect to the norm $|\cdot|_p$.

In fact, we have

$$|T_h^k(V_1) - T_h^k(V_2)|_p = e^{\lambda h}|\Psi(V_1 - V_2)|_p \le e^{\lambda h}|\Psi|_p|V_1 - V_2|_p \le e^{\lambda h}|\Psi|_p|V_1 - V_2|_p \tag{6.2}$$

proving that T_h^k is a contraction on the whole of $\mathbb{R}^q$ since $e^{\lambda h} < 1$. ◻

Remark 9. Comparing $(\mathrm{SP_h})$ and $(\mathrm{SP_h^k})$ it is easy to see that (4.5) ensures that the value $v_h^k(x_j + h\Phi_f(x_j))$ may be expressed as

$$v_h^k(x_j + h\Phi_f(x_j)) = \sum_{m\in Q} v_m \psi_m(x_j + h\Phi_f(x_j)).$$

The rate of convergence of the fully discretized solution is given by the following theorem (which is an extension of the result proved in [F]).

THEOREM 6.2. *Let* (h1), (h2), (2.4)–(2.7), (4.1), *and* (4.3) *hold true. Moreover, assume that* $v \in W^{1,\infty}(\mathbb{R}^N)$ *and that* $\|v - P_k v\|_\infty \le E(k)$*. Then, the following estimates hold for some positive constant* C*:*

(i) *If* $|\Psi|_\infty \le 1$ *then, for any* $h < h_0$*,*

$$\left\|v_h^k - v\right\|_\infty \le C\left(h^p + \frac{1}{h}E(k)\right). \tag{6.3}$$

(ii) *Let* $|\Psi|_2 \le 1$ *and* μ_2 *be a positive constant such that* $|M(\psi)|_2 \le \mu_2 k^N$ *for any* k*. Then, for any* $h < h_0$*,*

$$\|v_h^k - v\|_2 \le C\left(h^p + \frac{1}{h}E(k)\right). \tag{6.4}$$

Proof. The results can be obtained by passing to the limit for $\bar t \to \infty$ in (4.7) and (4.8), noting that the estimation is independent of $\bar t$ when $\lambda < 0$.

Remark 10. The global error estimates for the fully discrete approximation may be obtained plugging into (4.7)–(4.8) and (6.3), (6.4) the particular expression for the interpolation error $E(k)$, which in turn depends on the basis functions ψ_j and on the regularity of the solution. For example, assume a piecewise linear space discretization. For this set of basis functions we have

$$\sum_{j\in Q} |\psi_j(x)| \equiv 1$$

and, hence, $|\Psi|_\infty = 1$. Moreover, if $v \in W^{r,\infty}(\Omega)$, $(r = 1, 2)$, then $E(k) \le Ck^r$. Using (4.9), we get

$$\left\|v_h^k(\bar t) - v(\bar t)\right\|_\infty \le C\left(h^p + \frac{k^r}{h}\right). \tag{6.5}$$

Note that the error related to the space discretization depends inversely on the time step h, and therefore there exists an "optimal relationship" between the two steps, this being characterized by the same order of convergence of the two terms in the right-hand side of (6.5). This condition implies

$$k \sim h^{p+1/r}$$

and the global convergence rate is p with respect to the time step, and $rp/p+1$ with respect to the space step. Such analysis (which might be carried out in a similar way for the other cases, see [FFM]) shows that the relationship between h and k depends on p and r. It should also be noted that as soon as $p+1>r$ the ratio $h/k=\Delta t/\Delta x$ tends to infinity so that the formal Courant–Friedrichs–Lewy (CFL) condition is violated. However, the scheme does not depend on a fixed number of nodes as in the finite differences approximations and the stencil varies with the vector field (growing with its modulus) so that the numerical domain of dependence of the solution contains its domain of dependence. The CFL condition is then satisfied.

Remark 11. In order to ensure a physically meaningful behavior of the approximate solution, it is sometimes required that the (approximate) trajectories of (1.1) starting from different mesh points do not intersect each other. Carrying out the calculations as in [SP, Appendix], we obtain the sufficient condition

$$h < \frac{1}{|J_f|}$$

(where J_f is the Jacobian matrix of the vector field f), which ensures (at least asymptotically as $h \to 0$) that trajectories do not intersect, independently of the Courant number.

Remark 12. In case we are unable to prove either of the stability conditions $|\Psi|_\infty \le 1$, $|\Psi|_2 \le 1$, we can give an estimate of the local truncation error

$$L_{h,k}(x) := \frac{1}{h}[v(x,h) - v_h^k(x,h)].$$

Assuming v_0 is a smooth function, $L_{h,k}(x)$ can be computed working as in (4.9)–(4.11). We obtain

$$\begin{aligned} |L_{h,k}(x)| &\le \frac{1}{h}\Big[|V^1 - W^1|_\infty \sum_{j\in Q} |\psi_j(x)| \; + E(k)\Big] \\ &\le \frac{1}{h}\Big[(C(1+L_v)h^{p+1} + E(k)) \sum_{j\in Q} |\psi_j(x)| + E(k)\Big] \\ &\le C \sum_{j\in Q} |\psi_j(x)| \left(h^p + \frac{1}{h}E(k)\right). \end{aligned}$$

Note that the term $\sum_j |\psi_j(x)|$ depends on k for polynomial bases and is constant for finite element bases.

7. Boundary conditions. The theory exposed so far applies for $\Omega = \mathbb{R}^N$, or when $\Omega \subset \mathbb{R}^N$ and the "invariance" condition (4.1) is satisfied. Now we want to sketch how to adapt all the previous results to the more general case in which a boundary condition is imposed on the inflow portion of the boundary, this being defined as

$$\Gamma_{in} := \{x \in \partial\Omega : f(x)\cdot\nu(x) > 0\}.$$

We will, therefore, couple (EP), (SP) with the Dirichlet boundary condition

$$v(x,t) = b(x), \qquad \forall x \in \Gamma_{in},\ t \in [0,T]. \tag{7.1}$$

In terms of characteristics, this amounts to saying that the solution $y(x,t)$ of (1.1) may have a finite exit time from Ω. More precisely, we will set

$$\theta(x) := \inf\{t \geq 0 : y(x,t) \in \mathbb{R}^N \setminus \Omega\}.$$

The representation formulae (1.4), (1.5) may be changed accordingly, obtaining

$$v(x,t) = \int_0^{t\wedge\theta(x)} e^{\lambda s} g(y(s))ds + e^{\lambda(t\wedge\theta(x))} v_0(y(t)) \tag{7.2}$$

where $a \wedge b = \min(a,b)$, and t can be finite or $+\infty$.

Turning to the discrete approximation, it becomes necessary to determine the first point at which the discrete trajectory defined by (2.2) crosses the boundary of Ω. For simplicity, we will assume that Ω is defined by means of a continuous function $\varphi : \mathbb{R}^N \to \mathbb{R}$ such that

$$\begin{cases} \varphi(x) < 0 & \text{for } x \in \Omega, \\ \varphi(x) = 0 & \text{for } x \in \partial\Omega, \\ \varphi(x) > 0 & \text{for } x \notin \Omega, \end{cases} \tag{7.3}$$

so that given a positive parameter h, we can define a variable step $\eta : \Omega \to \mathbb{R}_+$ as follows:

$$\eta(x) := \min\{\inf\{t \in \mathbb{R}_+ : x + t\Phi_f(x,t) \in \mathbb{R}^N \setminus \Omega\}, h\}. \tag{7.4}$$

When $x + h\Phi_f(x,h)$ is outside Ω, this time step can be computed as the first t such that

$$\varphi(x + t\Phi_f(x,t)) = 0. \tag{7.5}$$

We expect it to be the only solution of (7.5) in $[0,h]$, at least for h small enough and "nonpathological" sets Ω.

Once computed $\eta(x)$ we modify the discrete scheme introducing the variable step. Set $Q = Q_i \cup Q_b$ such that $\eta(x_j) = h$ for $j \in Q_i$, $\eta(x_j) < h$ for $j \in Q_b$. Then, ($\mathrm{EP}^{\mathrm{k}}_{\mathrm{h}}$), ($\mathrm{SP}^k_h$) become

$$(\mathrm{EP}^{\mathrm{k}\prime}_{\mathrm{h}}) \quad \begin{cases} v_j^n = h\Phi_g(x_j,h) + e^{\lambda h} \sum_{m\in Q} v_m^{n-1} \psi_m(x_j + h\Phi_f(x_j,h))\ , \quad j \in Q_i, \\ v_j^n = \eta(x_j)\Phi_g(x_j,\eta(x_j)) + e^{\lambda\eta(x_j)} b(x_j + \eta(x_j)\Phi_f(x_j,\eta(x_j)))\ , \quad j \in Q_b, \\ v_j^0 = v_0(x_j), \end{cases}$$

$$(\mathrm{SP}^{\mathrm{k}\prime}_{\mathrm{h}}) \quad \begin{cases} v_j = h\Phi_g(x_j,h) + e^{\lambda h} \sum_{m\in Q} v_m \psi_m(x_j + h\Phi_f(x_j,h))\ , \quad j \in Q_i, \\ v_j = \eta(x_j)\Phi_g(x_j,\eta(x_j)) + e^{\lambda\eta(x_j)} b(x_j + \eta(x_j)\Phi_f(x_j,\eta(x_j)))\ , \quad j \in Q_b. \end{cases}$$

Note that this change will modify the discrete operator at the points x_j such that $\eta(x_j) < h$. In practice, this can affect only the nodes belonging to a neighborhood of the boundary (the radius of this neighborhood depending on both h and the vector field f). Moreover, the structure of the scheme is still given by (4.6), except for the following changes in the definition of the matrix Ψ and of the vector G:

$$\Psi := (\psi_{ij}) \ , \quad G := (g_i), \tag{7.6}$$

$$\begin{cases} \psi_{ij} = \psi_j(x_i + h\Phi_f(x_i, h)) \ , & i \in Q_i, \\ \psi_{ij} = 0 \ , & i \in Q_b, \end{cases}$$

$$\begin{cases} g_i = h\Phi_g(x_i, h) \ , & i \in Q_i, \\ g_i = \eta(x_i)\Phi_g(x_i, \eta(x_i)) + b(x_i + \eta(x_i)\Phi_f(x_i, \eta(x_i))) \ , & i \in Q_b. \end{cases}$$

By a straightforward adaptation of Theorem 6.1 one can prove the following theorem.

THEOREM 7.1. *Assume* (h1), (h2), (4.1), *and* (4.3). *Then* $(\mathrm{SP}_\mathrm{h}^{\mathrm{k}\prime})$ *has a unique solution.*

Remark 13. A direct proof of convergence of the time-discretized variable step scheme to the viscosity solution of (SP) can be obtained adapting the arguments in Theorem 3.2 of the paper by Camilli, Falcone, Lanucara, and Seghini [CFLS].

It is also possible to extend Theorems 4.1, 4.2, and 6.2 to this version of the scheme. The proofs of sections 4 and 5 account for the estimation in nodes where $\eta(x) = h$. When $\eta(x) < h$, a crucial point is to prove the order of convergence of $\eta(x)$ to the true exit time $\theta(x)$. To preserve the order of the scheme, it is necessary that $|\eta(x) - \theta(x)| = O(h^{p+1})$; this requires in general that the vector field f should not be tangent to the boundary of Ω.

We conclude making some remarks on the practical implementation of the variable-step technique. As we observed, it is not necessary to compute the variable step at each node of the grid. This will only be necessary for nodes in the neighborhood of the boundary, and in this case $\eta(x)$ is computed by solving (7.5). The search for the zero is local so that a Newton method will converge in a few steps provided f and φ are regular enough. The variable step is computed just once at the beginning and as a result the performance of the variable step algorithm is very close to that of the fixed-step algorithm.

Treating periodic boundary conditions in $(\mathrm{EP}_\mathrm{h}^{\mathrm{k}\prime})$, $(\mathrm{SP}_\mathrm{h}^{\mathrm{k}\prime})$ is even simpler. If we solve the problem in the set $\Omega = (0, P)$, then we can assign a value to each point outside Ω by the relationship

$$v_h^k(P + x, t) = v_h^k(x, t). \tag{7.7}$$

8. Numerical tests. We present in this section the results of some double precision numerical test to show behavior and efficiency of the schemes. The time discretization is carried out by means of either a second-order (Heun) or a fourth-order (Runge–Kutta) scheme, whereas the space discretizations used are P_1, P_2 (with a uniform grid) or polynomial (with a Chebyshev grid). For polynomial approximations, the space step refers to finite elements approximations with the same number of nodes (note that global polynomial discretizations are not covered by the stability analysis of section 5, but they have been included in order to show the robustness of the schemes).

We also present some numerical evaluation of the order of convergence with respect to the space step k. Note that negative orders of convergence appear when the error increases for decreasing discretization steps.

TABLE 1a
L^∞ errors for Example 1.

Runge–Kutta/P_1			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$2.1 \cdot 10^{-2}$	$1.5 \cdot 10^{-2}$	$2.6 \cdot 10^{-2}$
$k = 0.05$	$2.0 \cdot 10^{-2}$	$1.4 \cdot 10^{-3}$	$7.0 \cdot 10^{-3}$
$k = 0.015$	$2.0 \cdot 10^{-2}$	$1.3 \cdot 10^{-4}$	$1.5 \cdot 10^{-3}$
$k = 0.005$	$2.0 \cdot 10^{-2}$	$1.5 \cdot 10^{-5}$	$1.4 \cdot 10^{-4}$

Runge–Kutta/Polynomials			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$2.0 \cdot 10^{-2}$	$1.9 \cdot 10^{-6}$	$1.8 \cdot 10^{-10}$
$k = 0.05$	$2.0 \cdot 10^{-2}$	$1.9 \cdot 10^{-6}$	$1.8 \cdot 10^{-10}$

TABLE 1b
Orders in L^∞ errors for Example 1.

Runge–Kutta/P_1		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ / $k = 0.05$	2.1587	1.1944
$k = 0.015$ / $k = 0.005$	1.9656	2.1587

Example 1. *Stationary equation with smooth solution.*
We consider here the stationary problem in $\Omega = (-1, 1)$:

$$-v(x) - xv_x(x) + x^2 = 0$$

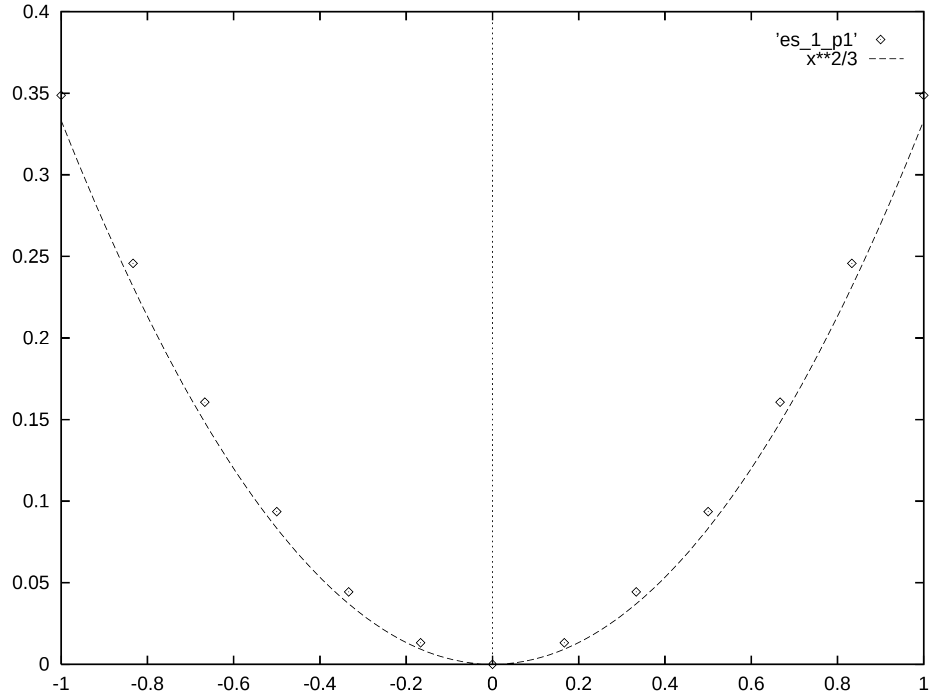
whose exact solution is given by

$$v(x) = \frac{x^2}{3}.$$

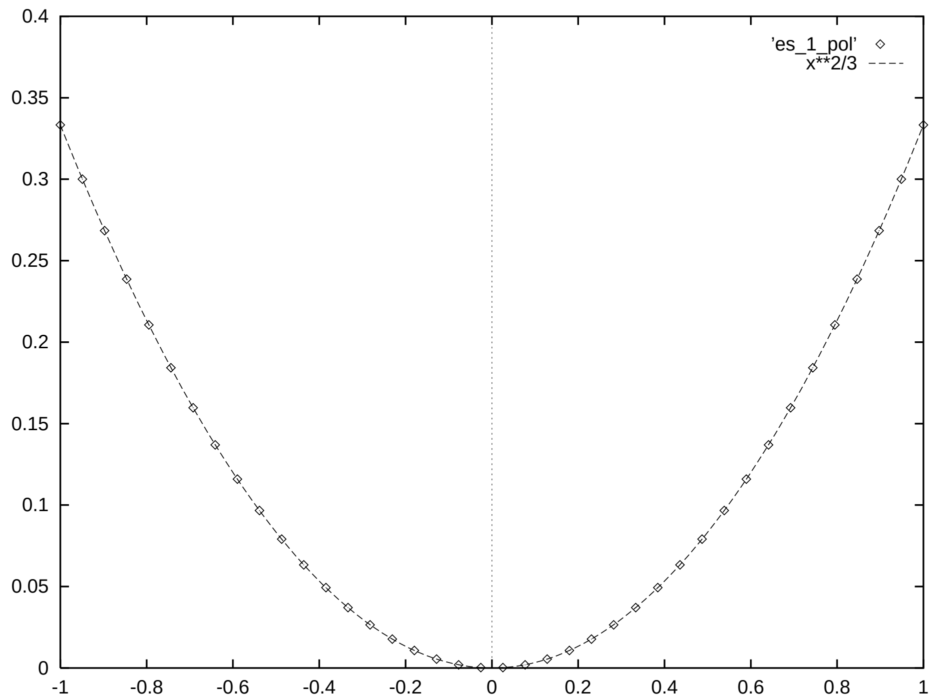
Table 1a shows the fully discrete approximation obtained by P_1 and polynomial basis functions. Note in the table that the accuracy of polynomial approximation is independent of the space step. This confirms the results of the convergence analysis, since for this example the interpolation error is zero, and hence the global error is only due to time discretization. As a consequence the order of convergence (with respect to k) is zero.

The coupling between Runge–Kutta and P_1 finite elements exhibit a second-order convergence with respect to the space step (see Table 1b). In fact, the error depends on the contribution of two terms. The first term, related to the time discretization, in this case is $O(h^4)$. The second term, related to the space discretization, is $O(k^2)/h$. For h decreasing, the first term becomes negligible and the order tends to 2. For the same reason, in this and in the following examples, the most accurate numerical estimate of the order is obtained by the last number in the rightmost column.

Figure 2 compares the exact solution with the P_1 and polynomial approximations with $h = 0.1$, $k = 0.15$.



(a)



(b)

FIG. 2. *Solutions for Example* 1*:* (a) *exact vs. Runge–Kutta/P_1 scheme,* (b) *exact vs. Runge–Kutta/polynomial scheme.*

Table 2a
L^∞ errors for Example 2.

Heun/P_1			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$7.5 \cdot 10^{-2}$	$1.4 \cdot 10^{-1}$	$2.0 \cdot 10^{-1}$
$k = 0.05$	$8.7 \cdot 10^{-2}$	$2.7 \cdot 10^{-2}$	$1.0 \cdot 10^{-1}$
$k = 0.015$	$9.5 \cdot 10^{-2}$	$7.0 \cdot 10^{-3}$	$4.3 \cdot 10^{-2}$
$k = 0.005$	$9.6 \cdot 10^{-2}$	$4.2 \cdot 10^{-3}$	$1.1 \cdot 10^{-2}$

Heun/P_2			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$9.4 \cdot 10^{-2}$	$5.8 \cdot 10^{-2}$	$5.0 \cdot 10^{-2}$
$k = 0.05$	$7.9 \cdot 10^{-2}$	$2.4 \cdot 10^{-2}$	$3.2 \cdot 10^{-2}$
$k = 0.015$	$7.9 \cdot 10^{-2}$	$2.4 \cdot 10^{-2}$	$1.1 \cdot 10^{-2}$
$k = 0.005$	$7.9 \cdot 10^{-2}$	$2.8 \cdot 10^{-3}$	$4.0 \cdot 10^{-3}$

Heun/Polynomials			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$9.2 \cdot 10^{-2}$	$4.6 \cdot 10^{-2}$	$4.2 \cdot 10^{-2}$
$k = 0.05$	$8.3 \cdot 10^{-2}$	$1.4 \cdot 10^{-2}$	$9.1 \cdot 10^{-3}$

Table 2b
Orders in L^∞ errors for Example 2.

Heun/P_1		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ / $k = 0.05$	1.4980	0.6309
$k = 0.015$ / $k = 0.005$	0.4649	1.2409

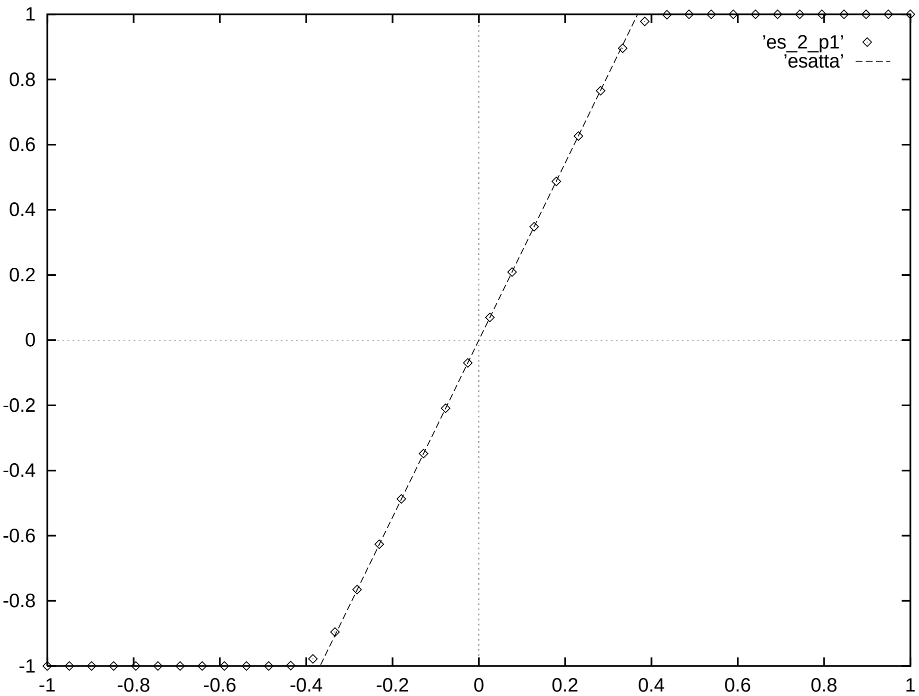
Heun/P_2		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ / $k = 0.05$	0.8031	0.4062
$k = 0.015$ / $k = 0.005$	0.8846	0.9207

Heun/Polynomials		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ / $k = 0.05$	1.0828	1.3921

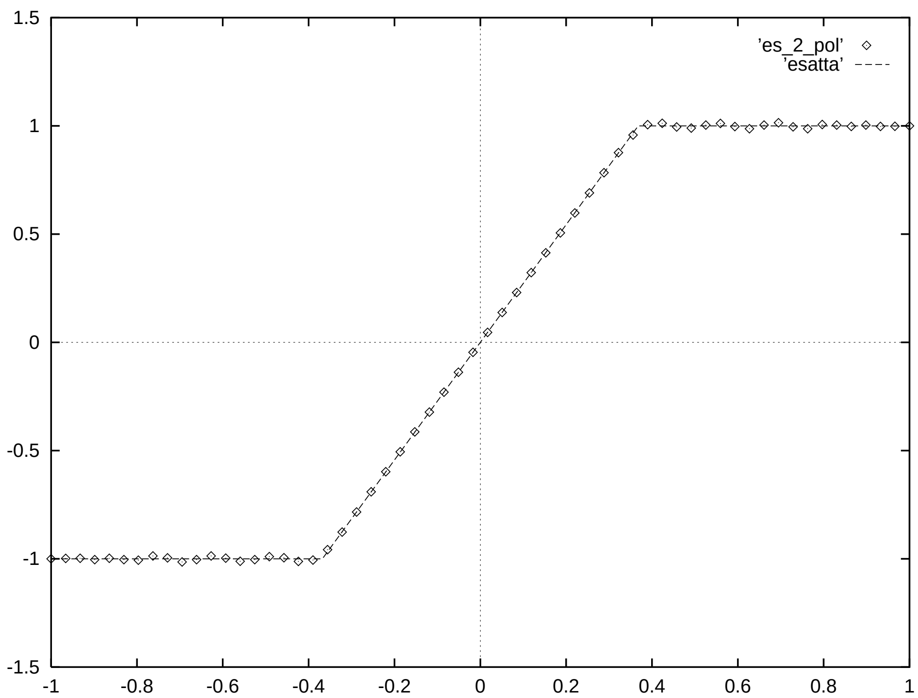
Example 2. *Evolution equation with $W^{1,\infty}$ solution.*
Let us now consider the evolution equation

$$\begin{cases} \dfrac{\partial}{\partial t} v(x,t) = x v_x(x,t), \\ v(x,0) = v_0(x), \end{cases} \tag{8.1}$$

posed on $\Omega = (-1, 1)$, with boundary conditions $v(-1,t) = -1$, $v(1,t) = 1$ and with $v_0(x) = x$. This case requires taking into account the boundary conditions since the



(a)



(b)

FIG. 3. *Solutions for Example 2:* (a) *exact vs. Heun/P_1 scheme,* (b) *exact vs. Heun/polynomial scheme.*

Table 3a
L^∞ errors for Example 3.

Runge–Kutta/P_1			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$6.0 \cdot 10^{-3}$	$8.3 \cdot 10^{-2}$	$1.1 \cdot 10^{-1}$
$k = 0.05$	$2.5 \cdot 10^{-3}$	$9.3 \cdot 10^{-3}$	$4.9 \cdot 10^{-2}$
$k = 0.015$	$2.1 \cdot 10^{-3}$	$9.5 \cdot 10^{-4}$	$1.3 \cdot 10^{-2}$
$k = 0.005$	$2.1 \cdot 10^{-3}$	$1.3 \cdot 10^{-4}$	$1.0 \cdot 10^{-3}$

Runge–Kutta/P_2			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$3.6 \cdot 10^{-2}$	$4.0 \cdot 10^{-2}$	$5.4 \cdot 10^{-2}$
$k = 0.05$	$2.1 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	$7.8 \cdot 10^{-3}$
$k = 0.015$	$2.1 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	$1.5 \cdot 10^{-3}$
$k = 0.005$	$2.1 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	$1.5 \cdot 10^{-3}$

Runge–Kutta/Polynomials			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$1.7 \cdot 10^{-2}$	$1.3 \cdot 10^{-2}$	$3.5 \cdot 10^{-2}$
$k = 0.05$	$2.8 \cdot 10^{-3}$	$1.3 \cdot 10^{-3}$	$1.3 \cdot 10^{-3}$

Table 3b
Orders in L^∞ errors for Example 3.

Runge–Kutta/P_1		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ $k = 0.05$	1.9923	0.7360
$k = 0.015$ $k = 0.005$	1.8104	2.3347

Runge–Kutta/P_2		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ $k = 0.05$	3.3577	1.7611
$k = 0.015$ $k = 0.005$	2.0959	3.0091

Runge–Kutta/Polynomials		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ $k = 0.05$	2.0959	2.9974

exit time $\theta(x)$ is finite for any $x \neq 0$. The exact solution is

$$v(x,t) = \begin{cases} -1 & \text{if} \quad x < -e^{-t}, \\ v_0(e^t x) & \text{if} \quad -e^{-t} < x < e^{-t}, \\ +1 & \text{if} \quad x > e^{-t}, \end{cases} \tag{8.2}$$

and has two corners at the points $\pm e^{-t}$. The errors shown in Table 2a refer to the time $\bar{t} = 1$ and show that finite elements and polynomials have similar perfomances due to the low regularity of the solution. The main source of error is time discretization

Table 4a
L^∞ errors for Example 4.

Runge-Kutta/P_1			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$9.5 \cdot 10^{-3}$	$1.1 \cdot 10^{-1}$	$1.3 \cdot 10^{-1}$
$k = 0.05$	$2.4 \cdot 10^{-3}$	$2.2 \cdot 10^{-2}$	$5.8 \cdot 10^{-2}$
$k = 0.015$	$2.0 \cdot 10^{-3}$	$1.8 \cdot 10^{-3}$	$1.7 \cdot 10^{-2}$
$k = 0.005$	$2.0 \cdot 10^{-3}$	$1.9 \cdot 10^{-4}$	$2.5 \cdot 10^{-3}$

Runge–Kutta/P_2			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$2.1 \cdot 10^{-3}$	$3.3 \cdot 10^{-2}$	$5.2 \cdot 10^{-2}$
$k = 0.05$	$2.5 \cdot 10^{-3}$	$2.6 \cdot 10^{-3}$	$1.0 \cdot 10^{-2}$
$k = 0.015$	$2.0 \cdot 10^{-3}$	$1.0 \cdot 10^{-4}$	$8.1 \cdot 10^{-4}$
$k = 0.005$	$2.0 \cdot 10^{-3}$	$3.0 \cdot 10^{-6}$	$1.4 \cdot 10^{-5}$

Runge–Kutta/Polynomials			
step	$h = 1.0$	$h = 0.1$	$h = 0.01$
$k = 0.15$	$2.3 \cdot 10^{-2}$	$1.6 \cdot 10^{-2}$	$6.5 \cdot 10^{-2}$
$k = 0.05$	$2.2 \cdot 10^{-3}$	$4.2 \cdot 10^{-4}$	$3.6 \cdot 10^{-4}$

Table 4b
Orders in L^∞ errors for Example 4.

Runge–Kutta/P_1		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ / $k = 0.05$	1.4649	0.7346
$k = 0.015$ / $k = 0.005$	2.0466	1.7448

Runge–Kutta/P_2		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ / $k = 0.05$	2.3129	1.5006
$k = 0.015$ / $k = 0.005$	3.1918	3.6937

Runge–Kutta/Polynomials		
step	$h = 0.1$	$h = 0.01$
$k = 0.15$ / $k = 0.05$	3.3133	4.7296

on the first column, space discretization on the third one. Figure 3 compares the exact solution with the P_1 and polynomial approximations for $h = 0.1$, $k = 0.05$. The errors are localized near the discontinuity in the P_1 scheme. This does not happen for the polynomial scheme, due to Gibbs' oscillations. The order of convergence is approximately 1 as we can see from Table 2b looking at the last number in the rightmost column.

TABLE 5a
L^∞ errors for Example 5.

Heun/P_1			
step	$h = \pi/10$	$h = \pi/20$	$h = \pi/40$
$k = 0.1$	0.90	0.83	0.94
$k = 0.05$	0.64	0.79	0.85
$k = 0.025$	0.40	0.69	0.66

Heun/P_2			
step	$h = \pi/10$	$h = \pi/20$	$h = \pi/40$
$k = 0.1$	0.38	0.39	0.57
$k = 0.05$	0.36	0.17	0.20
$k = 0.025$	0.38	0.13	0.11

Runge–Kutta/P_2			
step	$h = \pi/10$	$h = \pi/20$	$h = \pi/40$
$k = 0.1$	0.22	0.39	0.57
$k = 0.05$	0.091	0.13	0.21
$k = 0.025$	0.058	0.074	0.096

TABLE 5b
L^∞ norms for Example 5.

Heun/P_1			
step	$h = \pi/10$	$h = \pi/20$	$h = \pi/40$
$k = 0.01$	0.010	0.058	0.050
$k = 0.05$	0.35	0.22	0.12
$k = 0.025$	0.74	0.57	0.38

Heun/P_2			
step	$h = \pi/10$	$h = \pi/20$	$h = \pi/40$
$k = 0.1$	0.74	0.61	0.43
$k = 0.05$	1.00	0.99	0.94
$k = 0.025$	1.00	1.00	1.00

Runge–Kutta/P_2			
step	$h = \pi/10$	$h = \pi/20$	$h = \pi/40$
$k = 0.1$	0.79	0.61	0.43
$k = 0.05$	1.01	0.99	0.93
$k = 0.025$	1.00	1.00	1.00

Example 3. *Evolution equation with $W^{2,\infty}$ solution.*
Consider again (8.1) with a different initial condition,

$$v_0(x) = \sin \frac{\pi x}{2}.$$

The exact solution is still given by (8.2) and is once (but not twice) differentiable. The error tables compare P_1, P_2, and polynomial approximations at $\bar{t} = 1$. Note that for this smoother solution, the advantage of higher-order space discretizations becomes apparent, but (as expected) the convergence rate is still similar in all cases as it can be seen in Table 3b. However, for a careful choice of k and h the coupling between Runge–Kutta and P_2 can give a convergence of order 3.

Example 4. *Evolution equation with $W^{3,\infty}$ solution.*
Consider the same data of Example 2, complemented with the initial condition

$$v_0(x) = \frac{3}{8}x^5 - \frac{5}{4}x^3 + \frac{15}{8}x.$$

TABLE 5c
Orders in L^∞ norm for Example 5.

Heun/P_1			
step	$h=\pi/10$	$h=\pi/20$	$h=\pi/40$
$k=0.1$ – $k=0.05$	0.4918	0.0712	0.1451
$k=0.05$ – $k=0.025$	0.6780	0.1952	0.3649

Heun/P_2			
step	$h=\pi/10$	$h=\pi/20$	$h=\pi/40$
$k=0.1$ – $k=0.05$	0.0780	1.1979	
$k=0.05$ – $k=0.025$	−.07803	0.3870	0.8624

Runge–Kutta/P_2			
step	$h=\pi/10$	$h=\pi/20$	$h=\pi/40$
$k=0.1$ – $k=0.05$	1.2735	1.5849	1.4405
$k=0.05$ – $k=0.025$	0.6498	0.8129	1.1292

The exact solution given by (8.2) is twice (but not three times) differentiable. Comparing the solution at time $\bar{t}=1$ with the numerical solutions related to P_1, P_2, and polynomial approximations (see Table 4a) one can observe the improvement in the convergence rate for higher-order space discretizations (see also Table 4b). As expected, the convergence rate is about 2 for P_1 finite elements, about 3 for P_2 finite elements and polynomials provided the ratio k/h is not too large.

Example 5. *Evolutive equation in* $\mathbb{R}^2$.

This test, known as the rotating cone problem, is a typical benchmark for the approximation of advection equations (see [SW], [SG], [TA]). It is intended to show the properties of the numerical algorithm with respect to the L^∞ error, the L^∞ norm of the solution, and the diffusion introduced by the scheme.

The advecting vector field is $f(x,y)=(y,-x)$ in $\Omega=(-0.5,0.5)\times(-0.5,0.5)$. The initial condition is

$$v(x,0)=\begin{cases} 1-\frac{|x-x_0|^2}{r^2} & \text{for } \ |x-x_0|\le r, \\ 0 & \text{elsewhere,} \end{cases}$$

so that the solution is Lipschitz continuous and its maximum value equals $v(x_0,0)=1$.

The errors have been computed at $\bar{t}=2\pi$, i.e., after one turn around the origin. Table 5a compares L^∞ errors for Heun/P_1, Heun/P_2, and Runge–Kutta/P_2 schemes, whereas Table 5b compares L^∞ norms of the approximate solutions for the same schemes (this being a measure of the viscosity introduced by the approximation). Figure 4 compares the exact solution with the Heun/P_1 and Runge–Kutta/P_2 approximations.

The second and third scheme show a considerable improvement both in the error and in the reduction of viscous effects. Although the overall convergence rate is of first order due to the low regularity of the solution (see Table 5c), the smoothing is

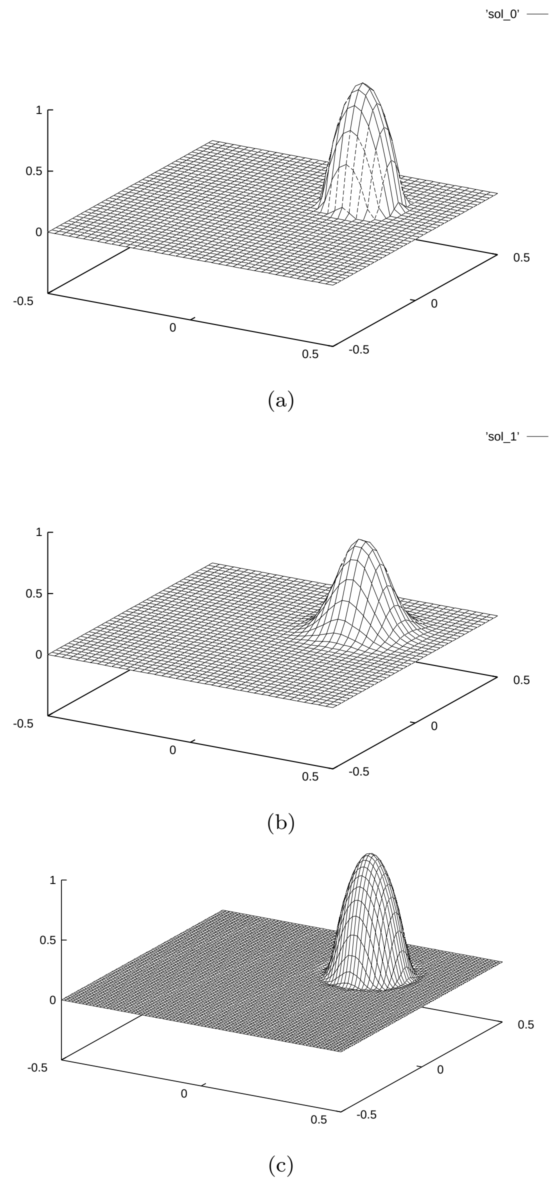


Fig. 4. *Solutions for Example* 4*:* (a) *exact,* (b) *Heun/P_1 scheme,* (c) *Runge–Kutta/P_2 scheme.*

localized near the discontinuities of the gradient for all schemes, and the qualitative behavior of the approximate solutions is very close to the exact one.

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